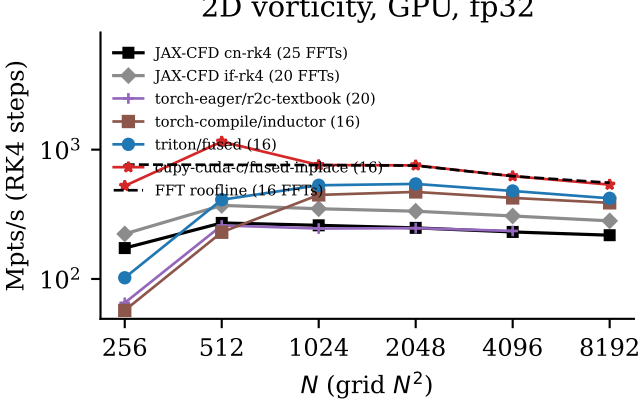
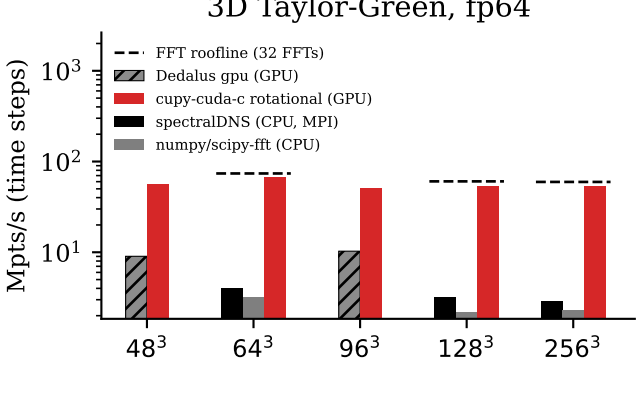

2D vorticity, GPU, fp32
JAX-CFD cn-rk4 (25 FFTs)
JAX-CFD if-rk4 (20 FFTs)
torch-eager/r2c-textbook (20)
torch-compile/inductor (16)
triton/fused (16)
cupy-cuda-c/fused-inplace (16)
FFT roofline (16 FFTs)
Mpts/s (RK4 steps)
10^3
10^2
256
512
1024
2048
4096
8192
N (grid N^2)


3D Taylor-Green, fp64
FFT roofline (32 FFTs)
Dedalus gpu (GPU)
cupy-cuda-c rotational (GPU)
spectralDNS (CPU, MPI)
numpy/scipy-fft (CPU)
Mpts/s (time steps)
10^3
10^2
10^1
48^3
64^3
96^3
128^3
256^3